

Hearing, balance, taste and smell

The written notes for the hearing, balance, taste and smell walkthrough. Week 5 competencies 20 to 23.

Step by step: where the signal travels

4 min

The path of sound into the cochlea

1. The outer ear funnels sound waves down the ear canal.
2. The tympanic membrane (eardrum) vibrates.
3. The malleus, incus and stapes carry the vibration across the middle ear.
4. The stapes pushes the oval window in and out.
5. The pressure wave travels along the vestibular duct, around the helicotrema at the tip, and back along the tympanic duct.
6. As the wave passes, the basilar membrane bows up and down; the round window bulges out and in to let the fluid move.

Hair cell transduction, step by step

1. At rest, about 10 percent of the hair cell channels are open, so the sensory neuron fires a steady tonic signal.
2. The basilar membrane moves, carrying the organ of Corti with it; the stereocilia bend against the tectorial membrane.
3. Bending toward the tallest stereocilium stretches the tip links and opens more mechanically gated cation channels.
4. K^+ flows in from the high- K^+ endolymph of the cochlear duct and the hair cell depolarizes.
5. Voltage-gated Ca^{2+} channels at the base open and more neurotransmitter is released.
6. The primary sensory neuron fires faster. Bending the other way closes channels, hyperpolarizes the cell, and slows or stops firing.

The path of the hearing signal to the cortex

1. Primary sensory neurons, with cell bodies in the spiral ganglion of the cochlea, carry action potentials in the cochlear branch of cranial nerve VIII.
2. They synapse in the cochlear nucleus in the medulla.

3. Signals go to the superior olivary nucleus on both sides; branches also go to the cerebellum.
4. They climb to the inferior colliculus in the midbrain.
5. They relay in the medial geniculate nucleus of the thalamus.
6. They reach the auditory cortex in the temporal lobe. Each ear sends information to both sides of the brain, so damage to one side of the brain rarely makes one ear deaf.

Sensing a head turn, step by step

1. The head turns; the bony skull and the canal walls turn with it.
2. The endolymph lags behind because of inertia.
3. In the ampulla, the drag of the endolymph bends the cupula of the crista opposite to the turn.
4. Hair cell cilia embedded in the cupula bend; bending toward the kinocilium depolarizes the cell, bending away hyperpolarizes it.
5. Transmitter release onto vestibular sensory neurons rises or falls.

The path of the balance signal

1. Hair cells release transmitter onto primary sensory neurons of the vestibular branch of cranial nerve VIII.
2. The neurons either synapse in the vestibular nuclei of the medulla or run without synapsing to the cerebellum, the main site for equilibrium processing.
3. From the vestibular nuclei, pathways go to the cerebellum, to the motor neurons that move the eyes (keeping the gaze steady as the head turns), and up through the reticular formation and thalamus to the cortex.

Sour taste, step by step

1. Food acid dissolves in saliva and H^+ reaches a type III (presynaptic) taste cell.
2. H^+ enters through a proton channel.
3. A K^+ leak channel closes, then Na^+ channels and Ca^{2+} channels open, depolarizing the cell.
4. Ca^{2+} entry triggers exocytosis of serotonin.
5. Serotonin excites the primary sensory (gustatory) neuron.

Sweet, bitter and umami taste, step by step

1. The tastant dissolves in saliva and binds a Gprotein-coupled receptor on a type II (receptor) cell: T1R for sweet and umami, T2R for bitter.
2. The G protein gustducin activates the phospholipase C pathway.
3. Ca^{2+} is released from intracellular stores.
4. Ca^{2+} opens Ca^{2+} -activated Na^{+} channels; Na^{+} entry depolarizes the cell.
5. ATP leaves the cell through the CALHM1 channel, not by exocytosis, and excites the primary sensory neuron. Support cells break down the leftover ATP.

The path of taste to the cortex

1. Gustatory neurons carry the signal in cranial nerves VII, IX and X.
2. They synapse in the medulla.
3. The signal relays in the thalamus.
4. It reaches the gustatory cortex in the insula. The brain reads the taste from which neuron populations respond most strongly.

The path of smell

1. An odorant dissolves in the mucus made by olfactory (Bowman's) glands.
2. It binds an olfactory receptor, a Gprotein-coupled receptor, on the cilia (the dendrites) of an olfactory sensory neuron, a bipolar neuron.
3. The G protein Golf raises cAMP; cAMP-gated cation channels open and the neuron depolarizes.
4. If the receptor potential reaches threshold, an action potential travels along the axon. The axons form cranial nerve I and pass through the bone to the olfactory bulb.
5. They synapse on secondary neurons in the olfactory bulb, many converging on each.
6. The olfactory tract carries the signal to the olfactory cortex, bypassing the thalamus, and to the amygdala and hippocampus of the limbic system.

3. The oval window pushes on the cochlear fluid; the round window bulges out to let the fluid move.
 4. The pressure wave moves the basilar membrane.
 5. Stereocilia bend toward the tallest; tip links pull mechanically gated channels open.
 6. Potassium enters from the high-potassium endolymph and the hair cell depolarizes.
 7. Voltage-gated calcium channels open at the base and glutamate is released onto auditory nerve fibers, which fire faster.
- Pitch is coded by place: high frequencies move the basilar membrane most near the oval window, where it is narrow and stiff; low frequencies move it most near the helicotrema, where it is wide and flexible.
 - The cochlea has three ducts: the vestibular duct (from the oval window), the cochlear duct (endolymph, high K^{+} , with the organ of Corti on the basilar membrane and the tectorial membrane above), and the tympanic duct (to the round window). The vestibular and tympanic ducts connect at the helicotrema.
 - Loudness is coded by rate: bigger movement, more transmitter, faster firing, more fibers recruited.
 - Hair cells make graded receptor potentials, not action potentials.

Hearing loss and the tuning fork tests

1 min

Conductive loss

Sound cannot get through the outer or middle ear (earwax, fluid, stiff ossicles).

Sensorineural loss

Damage to the cochlea or the auditory nerve.

Bone conduction

Vibration carried through the skull straight to the cochlea, bypassing the outer and middle ear.

	Weber (fork on the midline)	Rinne (bone, then air, same ear)
Normal	Heard equally or in the middle	Air longer than bone
Conductive loss in one ear	Louder in the affected ear	Bone longer than air in the affected ear
Sensorineural loss in one ear	Louder in the good ear	Air longer than bone, both shorter in the affected ear

Hearing

1 min

1. Sound, a pressure wave in air, vibrates the tympanic membrane.
2. The ossicles (malleus, incus, stapes) carry the vibration to the oval window. The eardrum is about 20 times the area of the oval window, so the same force on a smaller area gives a higher pressure, enough to move fluid instead of reflecting off it.

Weber alone shows the ears differ but not which kind of loss; Rinne in the ear Weber points to tells you.

Balance (equilibrium)

1 min

- Equilibrium has a dynamic component (movement through space) and a static component (head position). The brain combines the inner ear with proprioceptors and vision.
- Vestibular apparatus (membranous labyrinth): the utricle and saccule (otolith organs) and three semicircular canals that join the utricle. It is filled with endolymph, high in K⁺ and low in Na⁺, secreted continuously and drained into a dural venous sinus.
- Vestibular hair cells have a kinocilium at one side of the bundle. Bending toward it depolarizes the cell; bending away hyperpolarizes it. The cells are tonically depolarized at rest.

	Semicircular canals	Utricle and saccule (otolith organs)
Detects	Rotational acceleration	Linear acceleration and head position
Sensory structure	Crista in the ampulla: hair cells with cilia in a gel cupula spanning the ampulla	Macula: hair cells with cilia in the otolith membrane, topped with otoliths (calcium carbonate and protein)
How hairs bend	Endolymph lags behind and drags the cupula opposite the turn	Otoliths slide with gravity or acceleration, dragging the otolith membrane
Specifics	Horizontal: turning, shaking the head no. Superior: nodding yes, somersault. Posterior: tilting toward a shoulder, cartwheel	Utricle (horizontal macula): forward acceleration, deceleration, head tilt. Saccule (vertical macula): vertical forces, such as an elevator

- Steady spin: the endolymph catches up and the turning signal fades. Sudden stop: the fluid keeps moving, bending the cupula the other way, so you feel yourself turning. Dancers spot to avoid this.

- Positional vertigo: loose otoliths float into the semicircular canals, causing brief dizziness with changes in position.
- Meniere's disease: excess endolymph raises inner ear pressure, causing episodes of vertigo and nausea, tinnitus, low-tone hearing loss and a feeling of fullness in the ear.

Taste and smell

1 min

	Taste (gustation)	Smell (olfaction)
Receptor cell	Taste receptor cell: nonneural epithelial cell in a taste bud; each senses one taste	Olfactory sensory neuron: a bipolar neuron, replaced about every two months from stem cells
Stimulus	Five tastes: sweet, sour, salty, bitter, umami (glutamate); must dissolve in saliva	Odorants; must dissolve in mucus; each neuron has one receptor type
Transduction	Sour: H ⁺ enters type III cells, serotonin released. Sweet, bitter, umami: GPCRs and gustducin on type II cells, ATP released. Salt: Na ⁺ entry, mechanism in humans unclear	GPCR, Golf, cAMP, cAMP-gated cation channels
Route	CN VII, IX, X, medulla, thalamus, gustatory cortex in the insula	CNI, olfactory bulb, olfactory tract, olfactory cortex and limbic system, no thalamic relay

- Type I taste cells are support cells; type II are receptor cells (sweet, bitter, umami); type III are presynaptic cells (sour).
- What each taste signals: sour, H⁺; salty, Na⁺; sweet and umami, nutritious food; bitter, possible toxins.
- Spicy and cooling sensations come from TRP channels on nerve endings in the mouth, carried by the trigeminal nerve (CN V). Carbonation is sensed by sour cells.

- Much of flavor is smell. Anosmia (loss of smell) occurs with colds and COVID-19 and can be an early sign of Alzheimer's and Parkinson's.
- Smell's direct route to the amygdala and hippocampus links odors with emotion and memory.